

TIMEMIXER++: A General Time Series Pattern Machine for Universal Predictive Analysis

A multi-scale, multi-resolution framework for forecasting, classification, anomaly detection, and imputation

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1. Introduction

- Time series have overlapping periodicities.
- RNN/TCN: weak long-range capture.
- Transformers: struggle with overlap.
- Need multi-scale, multi-period modeling.
- Align with real-world multi-scale generation.

Goal: one model for diverse temporal structures.

Abstract

- Tasks: forecasting, classification, anomaly, imputation.
- Introduce TSPM: TIMEMIXER++.
- Idea: mix across time and frequency.
- Modules: MRTI, TID, MCM, MRM.
- State-of-the-art on eight tasks.



Contributions

- Convert series to multi-resolution time images.
- Dual-axis decomposition: seasonality and trend.
- Mix across scales and resolutions.
- SOTA on 8 tasks, 30 benchmarks.

8 Tasks

30 Benchmarks

27 Baselines

SOTA Results

2. Related Work

Time Series Models

- ARIMA, STL: trend/seasonal; limited nonlinearity.
- RNN/TCN: local focus; short dependencies.
- Transformers: long-range via attention.
- TimesNet, MLPs: 2D reshaping and simplicity.

Hierarchical Modeling

- Prior: moving averages then modeling.
- Limitation: rigid decomposition.
- Ours: dual-axis latent attention.
- Scales and resolutions interact hierarchically.

Positioning

Flexible latent decomposition

3. Architecture

- Encoder-only: input, MixerBlocks, output.
- Input: downsample to multi-scale set.
- MixerBlock: MRTI → TID → MCM → MRM.
- Output: scale heads and ensemble.

Figure 2 placeholder: Framework flow_{1D} → time images → mix → outputs

Design

From 1D to rich 2D

3. Multi-scale Time Series

- Input $x_0 \in \mathbb{R}^{T \times C}$.
- Downsample M times with stride 2.
- $X_{\text{init}} = \{x_0 \dots x^M\}$ with shorter lengths.

```
x_m = ConvStride2(x_{m-1})
```

Scales

Coarse to fine context

3.1 Input Projection

- Channel mixing at coarsest scale.
- Variate-wise self-attention across variables.
- Embed all scales to d_{model} .

```
[Q_M,K_M,V_M]=Linear(x^M); Channel-Attn
```

Cross-variable

Global integration

3.2 MixerBlock

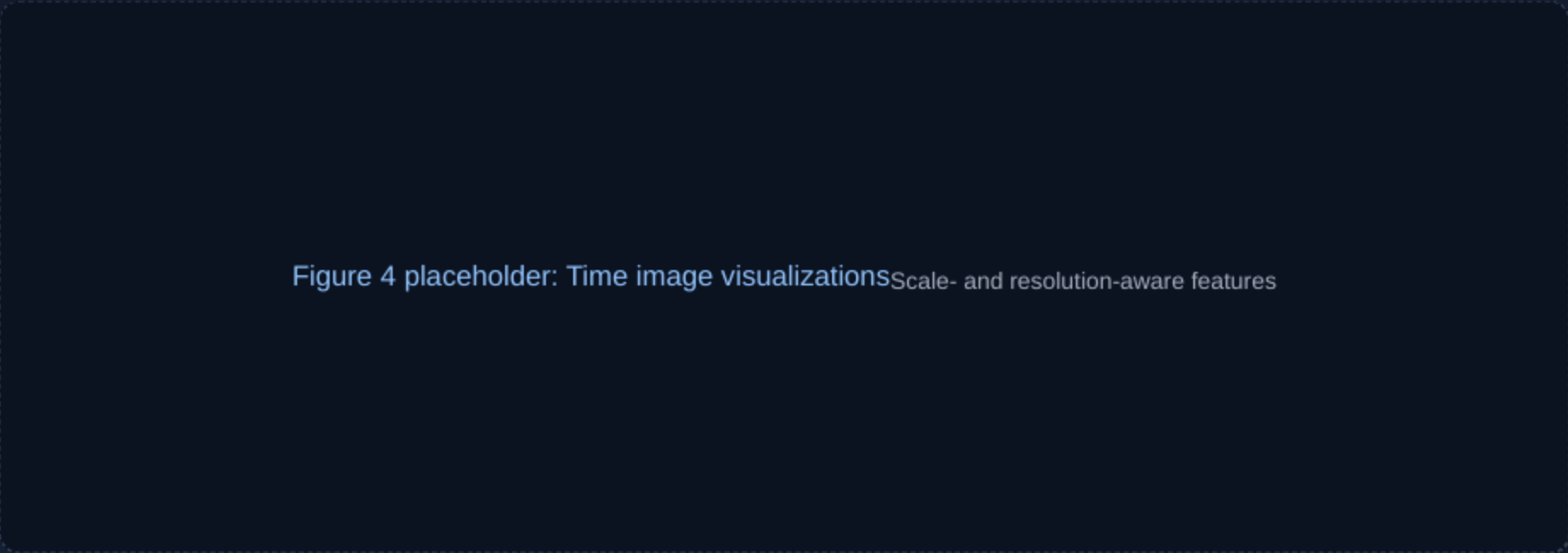
- LayerNorm and residual stack.
- MRTI: FFT-guided period selection.
- TID: dual-axis attention outputs s, t.
- MCM: seasonal bottom-up; trend top-down.
- MRM: amplitude-weighted fusion.

Block Flow

MRTI → TID → MCM → MRM

Representation Analysis

- Time images: original, seasonality, trend.
- Multi-period and time-varying trends.
- CKA: task-dependent layer similarity.



Insight

Adaptive depth patterns

4. Experiments: Summary

- Eight tasks and thirty benchmarks.
- Consistent SOTA versus twenty-seven baselines.
- Strong generalization and transfer ability.

Long-term Best MSE/MAE on eight sets.	M4 uni Lowest SMAPE, MASE, OWA.	PeMS multi Large MAE/MAPE reductions.
Imputation Lowest errors across masks.	Few-shot Wins with 10% training data.	Zero-shot Best ETT transfers.
Classify 75.9% accuracy overall.	Anomaly F1 87.47% across sets.	Takeaway One model wins broadly.

Scope

Single-model superiority

5. Conclusion

- Multi-resolution imaging and dual-axis decomposition.
- Mixing across scales and resolutions.
- SOTA over eight tasks and domains.

See appendices for details and limitations.

End

Universal pattern machine

Limitations

- Non-periodic spikes may be under-modeled.
- Sensitivity to top-K frequency selection.
- Compute overhead from 2D conversions.
- Large memory at high resolutions.

Caveats

Targets for future work